

# Empirical Diurnal Occupancy Profiles in Commercial and Residential Buildings: A Comparative Assessment of Measured Data and Code Baselines

Predictive building energy modeling, load matching, and grid-interactive efficient building operations depend heavily on the accuracy of diurnal occupancy schedules.<sup>1</sup> For decades, building simulationists relied on deterministic default schedules from standards like ASHRAE 90.1, the U.S. Department of Energy (DOE) Commercial Prototype Buildings, and the Canadian National Energy Code for Buildings (NECB).<sup>2</sup> However, these standardized schedules, often constructed from historical engineering judgment rather than empirical data, systematically overestimate occupancy peaks and fail to capture the stochastic, highly variable patterns of human presence.<sup>1</sup>

Through the widespread deployment of internet-of-things (IoT) sensors, mobile device tracking, and national time-use diaries, researchers have amassed extensive empirical databases that reveal profound gaps between code-compliance assumptions and actual building operations.<sup>8</sup> Highlighting these discrepancies reveals the structural variations in occupancy across four primary building typologies: residential dwellings, offices, retail spaces, and hotels.

## Residential Dwellings

The diurnal profile of residential occupancy, representing the fractional probability of occupant presence at home, is characterized by a high, stable nighttime plateau, morning departures, a prolonged midday trough, and an evening return phase.<sup>11</sup> Under weekday conditions, presence reaches its peak at approximately 3:00 AM, with a fractional value approaching 1.0, driven by sleeping behaviors localized within bedrooms.<sup>11</sup> Occupancy begins to steadily decline after 5:00 AM as individuals wake and prepare for departure.<sup>11</sup>

The weekday occupancy trough is reached between 11:00 AM and 4:00 PM, with presence rates dropping to a baseline range of 0.15 to 0.40 depending on the demographic profile, employment status, and household size of the occupants.<sup>11</sup> A steady evening ramp-up occurs between 5:00 PM and 8:00 PM as occupants return, leading to an active evening plateau dominated by living room and dining room presence.<sup>11</sup>

On weekends, the morning decline is significantly delayed and less pronounced, with occupants remaining asleep later and maintaining a much higher, stable daytime presence fraction.<sup>11</sup> Spatial analysis reveals that while dining room usage is restricted to a single evening dinner spike on weekdays, weekend curves exhibit distinct bi-modal spikes at lunch and dinner, indicating a higher propensity for eating at home.<sup>11</sup> Demographics dramatically shift these shapes; older populations (over 65 years of age) maintain highly similar weekday and weekend

profiles, showing elevated daytime presence and minimal absence throughout the week.<sup>11</sup> Standard energy modeling tools, such as those employing BEopt or simplified DOE residential prototype profiles, rely on static hourly fractional multipliers to estimate internal heat gains and domestic hot water loads.<sup>1</sup> When evaluated against empirical datasets—such as 12 to 14 years of longitudinal data from the American Time Use Survey (ATUS)—these standardized schedules exhibit deviations of up to 41% during specific diurnal periods.<sup>6</sup>

For example, reference schedules systematically underestimate midday occupancy for student and hybrid-working cohorts; empirical data shows that the actual probability of student presence at home during midday hours is double the rate projected by standard modeling assumptions.<sup>15</sup> This discrepancy distorts the predicted performance of residential demand-controlled ventilation (DCV) and smart thermostat schedules.<sup>6</sup>

Weekday occupancy schedules are heavily influenced by students' class schedules, where the average probability that students are home mid-day is twice that defined by the reference occupancy schedule.<sup>15</sup> Interestingly, the differences between the reference and actual occupancy profiles have produced a small change (~3%) in the savings predicted for occupancy-based heating and cooling controls in cold climates (4.3%–7.7%).<sup>15</sup> The potential savings was not sensitive to the variability among student occupancy schedules due to the high probability of occupancy for most hours of the day.<sup>15</sup>

The underlying data collection frameworks in North America and Europe introduce distinct modeling structures. North American residential profiles are heavily derived from the ATUS and the Residential Energy Consumption Survey (RECS).<sup>1</sup> In Europe, residential occupancy profiles are mapped using the Harmonised European Time Use Surveys (HETUS) framework, which coordinates time-diary data collection across the EU27 member states.<sup>16</sup>

Geopolitical differences in space standards greatly influence spatial-temporal occupancy densities. Per capita floor area is highest in Northern and Western Europe—exemplified by

Helsinki, where single-person households average 49  $\text{m}^2$  of floor area—and lowest in Central and Eastern Europe.<sup>17</sup> These spatial variations alter the localized thermal load density, meaning that European homes often exhibit lower internal occupant-driven heat gains per square meter than North American dwellings, which typically feature larger average home sizes but highly clustered spatial activities.<sup>17</sup>

## Office and Workplace Environments

Empirical workspace occupancy exhibits a distinct bi-modal distribution on weekdays, reflecting a morning arrival ramp-up, a midday lunch dip, an afternoon peak, and an evening departure ramp-down.<sup>2</sup> Morning arrivals commence steadily around 7:00 AM, climbing to a morning plateau by 9:30 AM.<sup>2</sup>

A highly consistent lunch trough occurs between 12:00 PM and 1:30 PM, where occupancy drops by approximately 30% to 50% relative to peak hours as occupants leave their immediate workspaces for dining areas or off-site locations.<sup>2</sup>

Occupancy recovers to its true daily peak at approximately 3:00 PM, followed by a rapid, steady decline after 5:00 PM.<sup>2</sup>

Between 7:00 PM and midnight, minor but highly consistent flat plateaus appear in empirical sensor data, which capture the structured, low-variability schedules of building maintenance and cleaning crews.<sup>2</sup>

A primary finding from longitudinal studies utilizing passive infrared (PIR) and desk-level sensors is that actual occupancy rarely, if ever, approaches the 100% design capacity assumed in engineering calculations.<sup>2</sup> In multi-tenant office buildings, empirical peak diversity factors for private offices range from only 0.50 to 0.55, and when analyzed on a month-to-month basis, these peak values fluctuate between 0.42 and 0.60.<sup>2</sup>

Open-plan offices demonstrate a higher peak presence but still plateau around 0.80.<sup>2</sup> ASHRAE Standard 90.1 Schedule AA default schedules, however, project a weekday peak occupancy of 95% to 100%.<sup>2</sup> This results in a massive occupancy overestimation of 46% for private offices and 12% for open-plan offices.<sup>2</sup>

Additionally, standard schedules project that the end-of-day occupancy decline begins at 5:00 PM, whereas empirical data shows that occupancy peaks at 3:00 PM and begins declining immediately thereafter.<sup>2</sup> This structural discrepancy leads to a substantial over-prediction of ventilation loads and equipment heat gains during late afternoon periods, contributing to the energy performance gap.<sup>2</sup>

Day-of-week variations are highly pronounced in empirical office data. Mondays consistently experience the highest peak occupancy of the workweek, whereas Fridays show the lowest occupancy levels, with occupants departing an average of 30 minutes earlier than on other weekdays.<sup>2</sup> Tuesdays, Wednesdays, and Thursdays exhibit no statistically discernible differences in occupancy levels.<sup>2</sup>

Seasonally, private office occupancy peaks in October, while the lowest occupancy profiles are recorded in August (driven by summer vacations) and November (due to extended Thanksgiving holiday travel in the United States).<sup>2</sup> Summertime occupancy does not show a shift in starting or leaving times, indicating that people generally maintain established work schedules regardless of the season.<sup>2</sup>

On days directly preceding federal holidays, occupancy curves closely mirror the early-departure patterns of Fridays.<sup>2</sup> On actual federal holidays, occupancy remains negligible, though minor spikes are occasionally captured due to intermittent cleaning operations or irregular working habits.<sup>2</sup> US Federal holidays show very low occupancy, which is likely due to occasional workers or cleaning/maintenance staff intermittently occupying a few offices.<sup>2</sup> The holidays ranked from the highest diversity factor peak to the lowest are Columbus Day, Veterans Day, Martin Luther King Jr. Day, Presidents Day, Labor Day, Independence Day, Christmas Day, Memorial Day, New Year's Day, and Thanksgiving.<sup>2</sup>

Energy compliance modeling in Canada under the National Energy Code of Canada for Buildings (NECB) and in the United States under ASHRAE 90.1 reveals differences in baseline calculations.<sup>4</sup> The NECB utilizes an energy-budget compliance path (comparing total modeled energy consumption), while ASHRAE 90.1 historically relies on an energy-cost budget method, comparing utility costs.<sup>21</sup>

Because the NECB is tailored to colder climates, its baseline assumptions for

space-conditioning equipment and heat recovery ventilation (HRV) thresholds differ.<sup>21</sup> In southern regions governed by ASHRAE 90.1, solar heat gain coefficient (SHGC) limits and cooling-driven occupancy schedules are highly emphasized, whereas Canadian models must prioritize thermal envelope performance and ventilation air preheating, where actual occupant density directly dictates the minimum ventilation load.<sup>21</sup>

## Retail and Shopping Environments

Retail environments require a strict analytical distinction between customer presence (footfall) and employee staffing schedules.<sup>23</sup> Customer occupancy curves are highly dynamic and non-uniform.<sup>23</sup> On weekdays, mall-based apparel and department stores experience a minor lunchtime spike (12:00 PM to 1:30 PM), a mid-afternoon trough, an after-school or late-afternoon rise, and a primary evening peak (5:30 PM to 7:30 PM) before closing.<sup>23</sup> Saturdays exhibit a single, massive, inverted U-shaped customer curve, peaking broadly between 1:00 PM and 4:00 PM.<sup>23</sup>

Staffing schedules, conversely, are typically static and highly structured, built around rigid 4-hour or 8-hour shift blocks to optimize labor budgets.<sup>23</sup> This structural mismatch leads to extreme, highly inefficient fluctuations in the staff-to-customer ratio throughout the trading day.<sup>23</sup>

For instance, a static retail schedule may produce a staff-to-customer ratio of 1:5 during the quiet weekday noon hour (overstaffed, resulting in unnecessary labor spend), a lean 1:45 ratio during the 3:30 PM after-school surge (severely understaffed, leading to customer checkout abandonment and lost conversion), and a 1:15 ratio by 5:00 PM.<sup>23</sup> On Saturdays, managers frequently overstaff shifts out of an anticipated fear of traffic, even when the actual localized workload does not justify the labor presence.<sup>26</sup>

This operational mismatch has severe psychological consequences: empirical studies indicate that excessive customer crowding causes negative emotional reactions, heightened workplace stress, and increased behavioral mistakes among frontline retail employees, while simultaneously reducing customer satisfaction and store dwell times.<sup>24</sup> Seventy-five percent of frontline service employees report negative emotions from crowding, noting that it produces stress, tension, and pressure, making it harder to interact positively with consumers.<sup>27</sup>

Furthermore, seventy-eight percent of respondents report behavioral effects, including a lack of concentration and an increased frequency of register entry errors.<sup>27</sup>

The standard retail occupancy schedule used in building energy simulations (Schedule CC) is characterized by a "historic non-evolution"; it has remained completely unchanged in standard compliance libraries from 1979 through current editions of ASHRAE 90.1, utilizing rigid, deterministic 10% incremental steps.<sup>3</sup>

Empirical evaluations comparing this standard schedule against real-time footfall data extracted from mobile device location platforms (such as Google Maps "popular times" across 13 major cities) reveal substantial deviations.<sup>3</sup> The mean peak values in actual buildings differ by 10% to 40% of the designed occupant capacity relative to standard assumptions.<sup>3</sup>

Furthermore, empirical data indicates that the traditional compliance categorization of retail

schedules into standard weekday, Saturday, and Sunday templates is fundamentally flawed, as local weather, urban morphology, and regional micro-locations completely reshape the diurnal signatures.<sup>3</sup>

Retail occupancy is highly volatile and seasonal, experiencing extreme, short-term surges during holiday promotional periods.<sup>29</sup> The primary seasonal peak occurs between late November and late December (the winter holiday shopping season), where customer occupancy levels can double or triple relative to standard spring or summer baselines.<sup>29</sup> These holiday peaks require rapid scaling of seasonal frontline staff, which often strains manager coaching capacities and exacerbates employee burnout due to high customer density and volatile interactions.<sup>31</sup>

frontline workers face unique pressures, with 61% having dealt with unruly customers (the highest rate across all industries) and 81% reporting burnout.<sup>31</sup> Only two-thirds of retail employees say staffing and resources are sufficient, highlighting the friction created when organizations attempt to meet high customer expectations with lean, inflexible operating models.<sup>31</sup>

## Hotel and Lodging Environments

Hotel buildings are highly complex, multi-functional environments where guest rooms and public spaces exhibit completely decoupled, out-of-phase diurnal occupancy profiles.<sup>32</sup>

Guest-room presence is dominated by overnight occupancy.<sup>34</sup> The typical empirical profile exhibits a high, stable nighttime plateau from 10:00 PM to 8:00 AM, a rapid morning ramp-down as guests check out or depart for daytime activities, and a deep, flat daytime trough between 9:00 AM and 5:00 PM.<sup>32</sup> During this daytime trough, rented rooms are frequently completely unoccupied, yet guests often maintain full control over individual thermostat overrides, leaving heating, ventilation, and air-conditioning (HVAC) systems and lighting running at maximum capacity.<sup>32</sup>

Lobbies, reception desk zones, and circulation corridors experience highly variable, transient occupancy that peaks during transition windows.<sup>35</sup> Lobby presence exhibits a sharp morning check-out spike between 10:00 AM and 12:00 PM, and a broad afternoon/evening check-in plateau between 4:00 PM and 7:00 PM.<sup>35</sup> Corridor and lobby maintenance labor, however, is frequently treated as a fixed service commitment, where identical cleaning and staffing coverage is maintained regardless of whether the hotel is operating at 31% or 89% occupancy, generating significant operational labor waste.<sup>33</sup>

On a Saturday at 89% occupancy with 340 guests moving through the lobby, standard coverage is appropriate; however, maintaining this identical staffing level on a Tuesday at 31% occupancy with only 118 guests creates a labor cost that does not scale with demand.<sup>33</sup>

The diurnal occupancy profile of a hotel is heavily dictated by its primary market segment.<sup>37</sup> Business-focused hotels, located near major urban centers or commercial hubs, exhibit elevated occupancy during the middle of the week (Tuesdays and Wednesdays), with occupancy ramping down significantly on Thursdays and remaining low throughout the weekend.<sup>37</sup> Leisure hotels and resorts demonstrate the exact inverse, with weekend (Friday and

Saturday night) room occupancy peaking close to 100%, while weekday presence drops to 50% or below, necessitating dynamic mid-week promotional pricing to smooth the demand curve.<sup>38</sup>

ASHRAE Standard 90.1 Appendix G and the DOE Commercial Prototype models represent hotel guest-room occupancy using Schedule FF, which assumes a static 60% daily average occupancy.<sup>40</sup> Because hotel guests do not pay individual electricity bills, their energy-use behaviors are highly unconstrained, representing a primary source of uncertainty in building performance prediction.<sup>41</sup>

To close this energy performance gap, recent building codes have transitioned from passive compliance to active control.<sup>34</sup> California Title 24 and ASHRAE 90.1-2019 mandate advanced guest-room occupancy-sensing controls.<sup>34</sup> These systems must be configured to automatically shut off mechanical ventilation to the guest room within 5 minutes of all occupants leaving, and adjust HVAC temperature setpoints by at least 5°F.<sup>34</sup> After 30 minutes of continuous vacancy, the temperature must be set back further (to at least 60°F for heating and 80°F for cooling), and all switched lighting and electrical receptacles must automatically shut off.<sup>34</sup>

Integrating specialized bathroom lighting controls also offers significant energy savings.<sup>44</sup> Installing wall-pack occupancy sensors set on an extra-long one-hour timeout prevents the lights from falsely turning off while a guest is showering, while ensuring that the bathroom luminaire does not burn indefinitely when unoccupied.<sup>44</sup>

From an HVAC design perspective, public areas require vastly different ventilation rates than guest rooms under ASHRAE Standard 62.1.<sup>45</sup> For example, a guest room requires a modest, low-variability outdoor air intake ( $V_{oz}$ ) based on a small occupant and floor area calculation<sup>45</sup>:

$$V_{oz} = (5 \text{ cfm/person} \times P_z) + (0.06 \text{ cfm/ft}^2 \times A_z)$$

In contrast, hotel lobbies must accommodate high, unpredictable transient populations ( $P_z$ ), requiring a significantly higher outdoor air intake calculated as<sup>45</sup>:

$$V_{oz,lobby} = (7.5 \text{ cfm/person} \times P_z) + (0.06 \text{ cfm/ft}^2 \times A_z)$$

This high occupant density makes lobbies prime candidates for demand-controlled ventilation (DCV) utilizing carbon dioxide ( $\text{CO}_2$ ) sensors to modulate outdoor air dampers based on real-time occupant counts, whereas guest rooms do not justify the cost of DCV due to their lower, relatively stable occupant densities.<sup>45</sup>

Empirical global hotel occupancy metrics indicate regional variations in tourism and travel dynamics. As of late 2022, average annual hotel occupancy rates were highest in Europe at 74%, followed by the Americas at 66%, the Middle East and Africa at 63%, and the Asia-Pacific region at 62%.<sup>39</sup>

Furthermore, research on thermal comfort standards shows that comfort expectations differ



by region.<sup>46</sup> Chinese lodging standards often underestimate comfort temperature thresholds, leading to summer overcooling, whereas ASHRAE recommendations in the United States permit wider temperature ranges than those inferred from empirical field evidence, occasionally resulting in localized thermal discomfort or unnecessary space-conditioning loads.<sup>46</sup>

## Diurnal Profiles Peak and Trough Hours Summary

The structured data below summarizes the empirical peak and trough hours, along with characteristic fractional multipliers, across weekdays, Saturdays, and Sundays for each investigated building use type.

| Building Use Typology                     | Day-Type | Peak Hour Range                        | Peak Fractional Range (Empirical)                       | Trough Hour Range                      | Trough Fractional Range (Empirical)                 |
|-------------------------------------------|----------|----------------------------------------|---------------------------------------------------------|----------------------------------------|-----------------------------------------------------|
| <b>Residential Dwellings</b> <sup>6</sup> | Weekday  | 10:00 PM – 5:00 AM                     | 0.90 – 1.00                                             | 11:00 AM – 4:00 PM                     | 0.15 – 0.40                                         |
|                                           | Saturday | 11:00 PM – 7:00 AM                     | 0.95 – 1.00                                             | 12:00 PM – 5:00 PM                     | 0.40 – 0.60                                         |
|                                           | Sunday   | 10:00 PM – 7:00 AM                     | 0.95 – 1.00                                             | 1:00 PM – 5:00 PM                      | 0.50 – 0.70                                         |
| <b>Office Environments</b> <sup>2</sup>   | Weekday  | 9:30 AM – 11:30 AM & 2:30 PM – 4:30 PM | 0.50 – 0.55 (Private Office)<br>0.75 – 0.80 (Open Plan) | 12:00 PM – 1:30 PM & 7:00 PM – 6:00 AM | 0.25 – 0.35 (Lunch Dip)<br>0.02 – 0.05 (Unoccupied) |
|                                           | Saturday | 10:00 AM – 2:00 PM                     | 0.05 – 0.10                                             | 4:00 PM – 8:00 AM                      | < 0.02                                              |
|                                           | Sunday   | 11:00 AM – 1:00 PM                     | < 0.05                                                  | 3:00 PM – 9:00 AM                      | < 0.01                                              |
| <b>Retail Customers</b> <sup>3</sup>      | Weekday  | 12:00 PM – 1:30 PM & 5:30 PM –         | 0.40 – 0.60                                             | 3:00 PM – 4:30 PM & 10:00 PM –         | 0.15 – 0.30 (Mid-Day Dip)                           |

|                                           |          |                                 |                                                           |                    |                     |
|-------------------------------------------|----------|---------------------------------|-----------------------------------------------------------|--------------------|---------------------|
|                                           |          | 7:30 PM                         |                                                           | 8:00 AM            | 0.00 (Store Closed) |
|                                           | Saturday | 1:00 PM – 4:00 PM               | 0.80 – 0.95                                               | 9:00 PM – 9:00 AM  | 0.00 (Store Closed) |
|                                           | Sunday   | 2:00 PM – 4:00 PM               | 0.60 – 0.75                                               | 6:00 PM – 11:00 AM | 0.00 (Store Closed) |
| <b>Retail Staff</b><br><sup>23</sup>      | Weekday  | 10:00 AM – 7:00 PM              | 0.50 – 0.70<br>(Static Shifts)                            | 9:00 PM – 8:00 AM  | 0.00 – 0.05         |
|                                           | Saturday | 10:00 AM – 7:00 PM              | 0.80 – 1.00<br>(Static Shifts)                            | 9:00 PM – 8:00 AM  | 0.00 – 0.05         |
|                                           | Sunday   | 11:00 AM – 5:00 PM              | 0.60 – 0.80<br>(Static Shifts)                            | 6:00 PM – 10:00 AM | 0.00                |
| <b>Hotel Guest Rooms</b><br><sup>32</sup> | Weekday  | 11:00 PM – 6:00 AM              | 0.75 – 0.90<br>(Business)<br><br>0.40 – 0.50<br>(Leisure) | 9:00 AM – 5:00 PM  | 0.10 – 0.20         |
|                                           | Saturday | 11:00 PM – 8:00 AM              | 0.90 – 1.00<br>(Leisure)<br><br>0.25 – 0.35<br>(Business) | 10:00 AM – 4:00 PM | 0.15 – 0.25         |
|                                           | Sunday   | 11:00 PM – 7:00 AM              | 0.60 – 0.80                                               | 11:00 AM – 5:00 PM | 0.20 – 0.30         |
| <b>Hotel Lobbies</b><br><sup>33</sup>     | Weekday  | 11:00 AM – 12:00 PM & 4:00 PM – | 0.60 – 0.75                                               | 1:00 AM – 6:00 AM  | < 0.05              |



|  |          |                                                 |             |                      |        |
|--|----------|-------------------------------------------------|-------------|----------------------|--------|
|  |          | 7:00 PM                                         |             |                      |        |
|  | Saturday | 10:00 AM –<br>1:00 PM &<br>3:00 PM –<br>6:00 PM | 0.80 – 0.95 | 2:00 AM –<br>7:00 AM | < 0.05 |
|  | Sunday   | 11:00 AM –<br>1:00 PM &<br>4:00 PM –<br>6:00 PM | 0.70 – 0.85 | 1:00 AM –<br>7:00 AM | < 0.05 |

## Methodological Suitability of Occupancy Data Sources

Evaluating building occupancy requires matching the research question with the appropriate data collection methodology. Person-level Time-Use Diaries (TUD), such as the American Time Use Survey (ATUS) or the Harmonised European Time Use Surveys (HETUS), provide self-reported, retrospective 24-hour records of individual activities and spatial contexts.<sup>1</sup>

These diaries are highly suited for residential dwellings, where occupancy is intrinsically tied to domestic, household-level schedules and member interactions.<sup>11</sup> TUD data allows researchers to construct stochastic Markov chain models that capture realistic home-presence durations and demographic correlations, such as age and income.<sup>6</sup>

However, TUDs are fundamentally unsuited for commercial, multi-occupant environments because they record individual trajectories rather than simultaneous spatial densities, fail to capture the physical capacity of specific buildings, and lack the real-time temporal resolution required for system control.<sup>19</sup>

Conversely, commercial and public typologies—such as offices, retail centers, and hotel lobbies—require population and footfall data sources.<sup>24</sup> Technologies such as passive Wi-Fi and Bluetooth Low Energy (BLE) tracking, thermal imaging sensors, and Time-of-Flight (ToF) cameras provide ground-truth, continuous spatial occupant counts.<sup>49</sup>

These methods are essential for capturing highly dynamic, non-uniform customer traffic in retail environments or checkout spikes in hotel lobbies.<sup>23</sup> They allow operators to calculate real-time staff-to-customer ratios or program demand-controlled ventilation (DCV) systems, which modulate outdoor air dampers based on real-time occupant counts.<sup>23</sup>

The primary limitation of these physical sensing technologies is their localized nature, which prevents broader demographic profiling, and their susceptibility to signal interference or line-of-sight constraints, whereas diary-based surveys provide statistically representative national datasets.<sup>1</sup>

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